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DATA MODULATION FOR ENHANCED LONG-RANGE MODE

Abstract

This disclosure describes systems, methods, and devices related to enhanced data modulation. A device may receive data to be transmitted over a wireless communication channel using a 20 MHz bandwidth and a plurality of 26-tone resource units (RUs). The device may determine a repetition type for the plurality of 26-tone RUs based on a transmission configuration, wherein the repetition type is selected from a first repetition and a second repetition. The device may map the data to the plurality of 26-tone RUs based on the repetition type, wherein the first repetition maps different data sets to subsets of the plurality of 26-tone RUs, each repeated three times, and the second repetition maps identical data across the plurality of 26-tone RUs. The device may transmit the data over the wireless communication channel using the plurality of 26-tone RUs.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application claims the benefit of U.S. Provisional Application No. 63/651,696, filed May 24, 2024, the disclosure of which is incorporated herein by reference as if set forth in full.

BACKGROUND

[0002] Wireless communication systems transmit data over shared frequency channels using various modulation and access techniques. As demands on range and reliability grow, there is a need for improved transmission methods that enhance performance without compromising compatibility.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. **1** is a network diagram illustrating an example network environment for enhanced data modulation, in accordance with one or more example embodiments of the present disclosure.

[0004] FIGS. **2-7** depict illustrative schematic diagrams for enhanced data modulation, in accordance with one or more example embodiments of the present disclosure.

[0005] FIG. **8** illustrates a flow diagram of a process for an illustrative enhanced data modulation system, in accordance with one or more example embodiments of the present disclosure.

[0006] FIG. **9** illustrates a functional diagram of an exemplary communication station that may be suitable for use as a user device, in accordance with one or more example embodiments of the present disclosure.

[0007] FIG. **10** illustrates a block diagram of an example machine upon which any of one or more techniques (e.g., methods) may be performed, in accordance with one or more example embodiments of the present disclosure.

[0008] FIG. **11** is a block diagram of a radio architecture in accordance with some examples.

[0009] FIG. **12** illustrates an example front-end module circuitry for use in the radio architecture of FIG. **11**, in accordance with one or more example embodiments of the present disclosure.

[0010] FIG. **13** illustrates an example radio IC circuitry for use in the radio architecture of FIG. **11**, in accordance with one or more example embodiments of the present disclosure.

[0011] FIG. **14** illustrates an example baseband processing circuitry for use in the radio architecture of FIG. **11**, in accordance with one or more example embodiments of the present disclosure.

DETAILED DESCRIPTION

[0012] The following description and the drawings sufficiently illustrate specific embodiments to enable those skilled in the art to practice them. Other embodiments may incorporate structural, logical, electrical, process, algorithm, and other changes. Portions and features of some embodiments may be included in, or substituted for, those of other embodiments. Embodiments set forth in the claims encompass all available equivalents of those claims.

[0013] Wi-Fi 8 (IEEE 802.11bn or ultra high reliability (UHR)) is the next generation of Wi-Fi and a successor to the IEEE 802.11be (Wi-Fi 7) standard. In line with all previous Wi-Fi standards, Wi-Fi 8 will aim to improve wireless performance in general, along with introducing new and innovative features to further advance Wi-Fi technology.

[0014] IEEE 802.11bn is likely to have an enhanced long-range mode (ELR). It is desirable that the required SNR of the extended long range (ELR) is lower than the legacy modulation and coding scheme (MCS) 0 by 6-9 decibels (dB). It is challenging to send the data and detect the packet arrival in such a low signal-to-noise ratio (SNR).
[0015] The preamble structure and physical layer (PHY) convergence protocol data unit (PPDU) structure for ELR were proposed. The modulation of the PPDU's data portion was also proposed. The power boosting of the components in the PPDU was proposed. In this disclosure, details about the data modulation and other aspects are provided.
[0016] Example embodiments of the present disclosure relate to systems, methods, and devices for data modulation of 802.11bn enhanced long-range mode.
[0017] For backward compatibility, it is desired to reuse the existing 26-tone, 52-tone RU, and 242-tone RU for ELR. To increase the received signal power, the frequency-domain repetition of the RU is applied. For reducing the peak to average power ratio (PAPR), bit masking (or phase rotation) and cyclic shift diversity (CSD) can be applied across the repeated RUs. The power of pilots may be boosted to enhance the carrier frequency offset (CFO) and phase tracking. Unlike the prior art, for some MCS, Repeating the same RU is not allowed over the whole band. Instead, multiple RUs with different data may be repeated across frequency.
[0018] The ELR improves the reliability of laptop connections such that the user experience gets improved.
[0019] For backward compatibility, it is desired to reuse the existing 26-tone, 52-tone Resource Units (RUs), and 242-tone RUs for Extended Long Range (ELR) transmission. A Resource Unit (RU) refers to a specific allocation of frequency tones in an Orthogonal Frequency-Division Multiple Access (OFDMA) system, enabling multiple users to transmit simultaneously. For example, a 26-tone RU allocates a narrow frequency band suitable for low-data-rate devices. To increase the received signal power, the frequency-domain repetition of the RU is applied. Frequency-domain repetition involves duplicating the same RU across different frequency locations to accumulate signal energy and enhance detection at the receiver. For example, a 52-tone RU may be repeated at two distinct frequency positions to improve signal strength in noisy environments. For reducing the peak-to-average power ratio (PAPR), bit masking (or phase rotation) and cyclic shift diversity (CSD) can be applied across the repeated RUs. Bit masking modifies specific bits of the signal to lower peak amplitudes, while phase rotation alters the signal phase to smoothen power fluctuations. CSD introduces phase shifts among RUs to increase robustness against channel fading. For instance, applying CSD to three repeated 26-tone RUs can enhance performance in a multipath-rich indoor setting. The power of pilots may be boosted to enhance the carrier frequency offset (CFO) and phase tracking. Pilots are known reference signals inserted within a transmission to aid the receiver in tracking frequency and phase deviations. Boosting pilot power improves the accuracy of CFO estimation and phase correction. For example, increasing pilot signal strength in a 242-tone RU improves synchronization for devices at extended distances. Unlike the prior art, for some Modulation and Coding Schemes (MCS), repeating the same RU is not allowed over the whole band. MCS defines the modulation format and error correction code rate, influencing data throughput and robustness. Instead, multiple RUs with different data may be repeated across frequency. For example, rather than repeating one RU three times, three different RUs carrying distinct data segments may be transmitted over separated frequency locations.
[0020] The ELR improves the reliability of laptop connections such that the user experience gets improved. By leveraging techniques like frequency repetition, PAPR reduction, and pilot boosting, ELR ensures more stable connections over longer distances, even in environments with interference or low signal strength. For example, in a congested office environment, ELR enables a laptop to maintain a video call without interruptions by dynamically adapting transmission parameters.

[0021] The above descriptions are for purposes of illustration and are not meant to be limiting. Numerous other examples, configurations, processes, algorithms, etc., may exist, some of which are described in greater detail below. Example embodiments will now be described with reference to the accompanying figures.
[0022] FIG. **1** is a network diagram illustrating an example network environment of enhanced data modulation, according to some example embodiments of the present disclosure. Wireless network **100** may include one or more user devices **120** and one or more access points(s) (AP) **102**, which may communicate in accordance with IEEE 802.11 communication standards. The user device(s) **120** may be mobile devices that are non-stationary (e.g., not having fixed locations) or may be stationary devices.
[0023] In some embodiments, the user devices **120** and the AP **102** may include one or more computer systems similar to that of the functional diagram of FIG. **9** and/or the example machine/system of FIG. **10**.
[0024] One or more illustrative user device(s) **120** and/or AP(s) **102** may be operable by one or more user(s) **110**. It should be noted that any addressable unit may be a station (STA). An STA may take on multiple distinct characteristics, each of which shape its function. For example, a single addressable unit might simultaneously be a portable STA, a quality-of-service (QoS) STA, a dependent STA, and a hidden STA. The one or more illustrative user device(s) **120** and the AP(s) **102** may be STAs. The one or more illustrative user device(s) **120** and/or AP(s) **102** may operate as a personal basic service set (PBSS) control point/access point (PCP/AP). The user device(s) **120** (e.g., **124**, **126**, or **128**) and/or AP(s) **102** may include any suitable processor-driven device including, but not limited to, a mobile device or a non-mobile, e.g., a static device. For example, user device(s) **120** and/or AP(s) **102** may include, a user equipment (UE), a station (STA), an access point (AP), a software enabled AP (SoftAP), a personal computer (PC), a wearable wireless device (e.g., bracelet, watch, glasses, ring, etc.), a desktop computer, a mobile computer, a laptop computer, an ultrabook™ computer, a notebook computer, a tablet computer, a server computer, a handheld computer, a handheld device, an internet of things (IoT) device, a sensor device, a PDA device, a handheld PDA device, an on-board device, an off-board device, a hybrid device (e.g., combining cellular phone functionalities with PDA device functionalities), a consumer device, a vehicular device, a non-vehicular device, a mobile or portable device, a non-mobile or non-portable device, a mobile phone, a cellular telephone, a PCS device, a PDA device which incorporates a wireless communication device, a mobile or portable GPS device, a DVB device, a relatively small computing device, a non-desktop computer, a “carry small live large” (CSLL) device, an ultra mobile device (UMD), an ultra mobile PC (UMPC), a mobile internet device (MID), an “origami” device or computing device, a device that supports dynamically composable computing (DCC), a context-aware device, a video device, an audio device, an A/V device, a set-top-box (STB), a blu-ray disc (BD) player, a BD recorder, a digital video disc (DVD) player, a high definition (HD) DVD player, a DVD recorder, a HD DVD recorder, a personal video recorder (PVR), a broadcast HD receiver, a video source, an audio source, a video sink, an audio sink, a stereo tuner, a broadcast radio receiver, a flat panel display, a personal media player (PMP), a digital video camera (DVC), a digital audio player, a speaker, an audio receiver, an audio amplifier, a gaming device, a data source, a data sink, a digital still camera (DSC), a media player, a smartphone, a television, a music player, or the like. Other devices, including smart devices such as lamps, climate control, car components, household components, appliances, etc. may also be included in this list.
[0025] As used herein, the term “Internet of Things (IoT) device” is used to refer to any object (e.g., an appliance, a sensor, etc.) that has an addressable interface (e.g., an Internet protocol (IP) address, a Bluetooth identifier (ID), a near-field communication (NFC) ID, etc.) and can transmit information to one or more other devices over a wired or wireless connection. An IoT device may have a passive communication interface, such as a quick response (QR) code, a radio-frequency identification (RFID) tag, an NFC tag, or the like, or an active communication interface, such as a

modem, a transceiver, a transmitter-receiver, or the like. An IoT device can have a particular set of attributes (e.g., a device state or status, such as whether the IoT device is on or off, open or closed, idle or active, available for task execution or busy, and so on, a cooling or heating function, an environmental monitoring or recording function, a light-emitting function, a sound-emitting function, etc.) that can be embedded in and/or controlled/monitored by a central processing unit (CPU), microprocessor, ASIC, or the like, and configured for connection to an IoT network such as a local ad-hoc network or the Internet. For example, IoT devices may include, but are not limited to, refrigerators, toasters, ovens, microwaves, freezers, dishwashers, dishes, hand tools, clothes washers, clothes dryers, furnaces, air conditioners, thermostats, televisions, light fixtures, vacuum cleaners, sprinklers, electricity meters, gas meters, etc., so long as the devices are equipped with an addressable communications interface for communicating with the IoT network. IoT devices may also include cell phones, desktop computers, laptop computers, tablet computers, personal digital assistants (PDAs), etc. Accordingly, the IoT network may be comprised of a combination of "legacy" Internet-accessible devices (e.g., laptop or desktop computers, cell phones, etc.) in addition to devices that do not typically have Internet-connectivity (e.g., dishwashers, etc.).

[0026] The user device(s) **120** and/or AP(s) **102** may also include mesh stations in, for example, a mesh network, in accordance with one or more IEEE 802.11 standards and/or 3GPP standards.

[0027] Any of the user device(s) **120** (e.g., user devices **124**, **126**, **128**), and AP(s) **102** may be configured to communicate with each other via one or more communications networks **130** and/or **135** wirelessly or wired. The user device(s) **120** may also communicate peer-to-peer or directly with each other with or without the AP(s) **102**. Any of the communications networks **130** and/or **135** may include, but not limited to, any one of a combination of different types of suitable communications networks such as, for example, broadcasting networks, cable networks, public networks (e.g., the Internet), private networks, wireless networks, cellular networks, or any other suitable private and/or public networks. Further, any of the communications networks **130** and/or **135** may have any suitable communication range associated therewith and may include, for example, global networks (e.g., the Internet), metropolitan area networks (MANs), wide area networks (WANs), local area networks (LANs), or personal area networks (PANs). In addition, any of the communications networks **130** and/or **135** may include any type of medium over which network traffic may be carried including, but not limited to, coaxial cable, twisted-pair wire, optical fiber, a hybrid fiber coaxial (HFC) medium, microwave terrestrial transceivers, radio frequency communication mediums, white space communication mediums, ultra-high frequency communication mediums, satellite communication mediums, or any combination thereof.

[0028] Any of the user device(s) **120** (e.g., user devices **124**, **126**, **128**) and AP(s) **102** may include one or more communications antennas. The one or more communications antennas may be any suitable type of antennas corresponding to the communications protocols used by the user device(s) **120** (e.g., user devices **124**, **126** and **128**), and AP(s) **102**. Some non-limiting examples of suitable communications antennas include Wi-Fi antennas, Institute of Electrical and Electronics Engineers (IEEE) 802.11 family of standards compatible antennas, directional antennas, non-directional antennas, dipole antennas, folded dipole antennas, patch antennas, multiple-input multiple-output (MIMO) antennas, omnidirectional antennas, quasi-omnidirectional antennas, or the like. The one or more communications antennas may be communicatively coupled to a radio component to transmit and/or receive signals, such as communications signals to and/or from the user devices **120** and/or AP(s) **102**.

[0029] Any of the user device(s) **120** (e.g., user devices **124**, **126**, **128**), and AP(s) **102** may be configured to perform directional transmission and/or directional reception in conjunction with wirelessly communicating in a wireless network. Any of the user device(s) **120** (e.g., user devices **124**, **126**, **128**), and AP(s) **102** may be configured to perform such directional transmission and/or reception using a set of multiple antenna arrays (e.g., DMG antenna arrays or the like). Each of the multiple antenna arrays may be used for transmission and/or reception in a particular respective direction or range of directions. Any of the user device(s) **120** (e.g., user devices **124**, **126**, **128**), and AP(s) **102** may be configured to perform any given directional transmission towards one or more defined transmit sectors. Any of the user device(s) **120** (e.g., user devices **124**, **126**, **128**), and AP(s) **102** may be configured to perform any given directional reception from one or more defined receive sectors.

[0030] MIMO beamforming in a wireless network may be accomplished using RF beamforming and/or digital beamforming. In some embodiments, in performing a given MIMO transmission, user devices **120** and/or AP(s) **102** may be configured to use all or a subset of its one or more communications antennas to perform MIMO beamforming.

[0031] Any of the user devices **120** (e.g., user devices **124**, **126**, **128**), and AP(s) **102** may include any suitable radio and/or transceiver for transmitting and/or receiving radio frequency (RF) signals in the bandwidth and/or channels corresponding to the communications protocols utilized by any of the user device(s) **120** and AP(s) **102** to communicate with each other. The radio components may include hardware and/or software to modulate and/or demodulate communications signals according to pre-established transmission protocols. The radio components may further have hardware and/or software instructions to communicate via one or more Wi-Fi and/or Wi-Fi direct protocols, as standardized by the Institute of Electrical and Electronics Engineers (IEEE) 802.11 standards. In certain example embodiments, the radio component, in cooperation with the communications antennas, may be configured to communicate via 2.4 GHz channels (e.g. 802.11b, 802.11g, 802.11n, 802.11ax), 5 GHz channels (e.g. 802.11n, 802.11ac, 802.11ax, 802.11be, 802.11bn, etc.), 6 GHZ channels (e.g., 802.11ax, 802.11be, 802.11bn, etc.), or 60 GHZ channels (e.g. 802.11ad, 802.11ay). 800 MHz channels (e.g. 802.11ah). The communications antennas may operate at 28 GHz and 40 GHz. It should be understood that this list of communication channels in accordance with certain 802.11 standards is only a partial list and that other 802.11 standards may be used (e.g., Next Generation Wi-Fi, or other standards). In some embodiments, non-Wi-Fi protocols may be used for communications between devices, such as Bluetooth, dedicated short-range communication (DSRC), Ultra-High Frequency (UHF) (e.g. IEEE 802.11af, IEEE 802.22), white band frequency (e.g., white spaces), or other packetized radio communications. The radio component may include any known receiver and baseband suitable for communicating via the communications protocols. The radio component may further include a low noise amplifier (LNA), additional signal amplifiers, an analog-to-digital (A/D) converter, one or more buffers, and digital baseband.

[0032] In one embodiment, and with reference to FIG. **1**, a user device **120** may be in communication with one or more APs **102**. For example, one or more APs **102** may implement an enhanced data modulation **142** with one or more user devices **120**. The one or more APs **102** may be multi-link devices (MLDs) and the one or more user device **120** may be non-AP MLDs. Each of the one or more APs **102** may comprise a plurality of individual APs (e.g., AP1, AP2, . . . , APn, where n is an integer) and each of the one or more user devices **120** may comprise a plurality of individual STAs (e.g., STA1, STA2, . . . , STAn). The AP MLDs and the non-AP MLDs may set up one or more links (e.g., Link1, Link2, . . . , Linkn) between each of the individual APs and STAs. It is understood that the above descriptions are for the purposes of illustration and are not meant to be limiting.

[0033] FIGS. **2-7** depict illustrative schematic diagrams for enhanced data modulation, in accordance with one or more example embodiments of the present disclosure.

[0034] The ELR PPDU structure of interest is shown in FIG. **2**. The packet detection is done at L-STF, which is power boosted by 3-6 dB, and the autoclassification, which identifies the ELR PPDU type, is done at ELR-STF. In some proposals, the autoclassification is done at a dedicated field, ELR-C, right before the ELR-STF. It is believed that the ELR-C and ELR-STF can be combined.

[0035] The Extended Long Range-Classification (ELR-C) contains information used for classification, for example, the ELR PPDU type or other things. Extended Long Range Short

Training Field (ELR-STF) is a specially designed training sequence within the preamble of ELR frames, enhancing signal detection, synchronization, and robustness under long-range and low SNR conditions. These components collectively enhance ELR's ability to support stable and reliable wireless communication over extended distances.

Joint Packet Detection and Autoclassification

[0036] Because it is hard to power boost the legacy-short training field (L-STF) by 6 or even 9 dB above the power of the data portion, it is unreliable to rely on the L-STF for the packet detection. Signals of L-STF and ELR-STF (or ELR-C) may be used jointly for packet detection. For example, a low threshold may be used for the L-STF detection to increase the sensitivity of the packet detection at the cost of increased false alarm. The false alarm is then silenced at the ELR-STF (or ELR-C) and the true alarm is also confirmed at the ELR-STF (or ELR-C). The ELR-STF or ELR-C may have a unique period other than 0.8/N microsecond (for integer N=1, 2, . . .) or have a special signal structure such as low PAPR sequences and sharp auto-correlation sequences like Golay or Zadoff-Chu sequence that can be easily identified. Because the pilot signals in L-SIG, RL-SIG, and U-SIG can be power boosted for enhancing the correction of the carrier frequency offset (CFO), the period of the ELR-STF (or ELR-C) can be longer than 0.8 microsecond like 1.6 microsecond for enhancing the reliability. By the joint signal processing on L-STF and ELR-STF (or ELR-C), the reliability of the packet detection gets improved. In addition, the autoclassification is also finished after the joint signal processing.

Different Repetitions of 26-tone RU

[0037] The RU positions in 20 MHz bandwidth are shown in FIG. **3**. Each RU like 26-tone RU has its own pilots and data subcarriers. For ELR, the power of the pilots may be boosted for enhancing the carrier frequency offset (CFO) and phase tracking. For backward compatibility, it is desired to reuse the existing RUs, i.e., 26-tone RU, 52-tone RU, 106-RU, and/or 242-tone RU.

[0038] In FIGS. **4**A-**4**B, two repetition types of 26-tone RU are defined, e.g., 9× repetition and 3× repetition. In 9× repetition, the same data are loaded onto the 9 26-tone RUs in the 20 MHz band. Namely, the same data get repeated **9** times. In 3× repetition, the 3 different sets of data are loaded onto the 9 26-tone RUs in the 20 MHz band and each set of data get repeated 3 times. For maximizing the frequency diversity gain, the repetition is desired to be in a round robin fashion as illustrated in FIG. **4**B.

[0039] The target data rates of ELR range from 1 Mbps to 8 Mbps. To achieve the target data rates, three combinations of code rate and modulation are added, e.g., code rate ⅔ with BPSK, code rate ¾ with BPSK, and code rate ⅔ with QPSK to the existing MCS in the **802**.**11** standards. Using the combinations of repetition number, code rate, and modulation, a set of data rates is generated between 0.8-7.5 Mbps for ELR as follows. Since the delay spread is likely to be large for ELR channels, 0.8 microsecond cyclic prefix (CP) may not be used especially for the uplink. Instead, 1.6 or 3.2 microseconds of CP should be used. For example, 1.6 microseconds may be used for both downlink and uplink, and 3.2 microseconds may be used for multiuser uplink. In the Table 1, 1.6 microseconds of CP are used.

TABLE-US-00001 TABLE 1 MCSs from 26-tone RU. Modulation Code Rate Repetition Number Date Rate (Mb/s) BPSK 1/2 9 0.8 BPSK 2/3 9 1.1 BPSK 3/4 9 1.2 QPSK 1/2 9 1.7 QPSK 2/3 9 2.2 QPSK 3/4 9 2.5 BPSK 1/2 3 2.5 BPSK 2/3 3 3.3 BPSK 3/4 3 3.7 QPSK 1/2 3 5.1 QPSK 2/3 3 6.6 QPSK 3/4 3 7.5

[0040] About 3-6 combinations may be picked with the target data rates, which have good performances in fading channels for BCC and LDPC codes, from the table above for ELR. It is likely that the lowest data rate of ELR may be about 1.5 Mbps instead of 1 Mbps. This lowest data rate excludes few options in the table. If code rate 3/4 with BPSK is not added to the existing MCS, few more options are excluded from the table. If two options have about the same data rate, the option with lower code rate may be selected for ELR.

Different Repetitions of 52-tone RU

[0041] There are four 52-tone RUs in 20 MHz as shown in FIG. **3**. For simplicity, the middle 26-tone RU straddling the DC may not be used. This may reduce the output power in exchange for simplicity. Two types of repetition are defined as shown in FIGS. **5**A-**5**B.

[0042] The target data rates of ELR range from 1 Mbps to 8 Mbps. To achieve the target data rates, three combinations of code rate and modulation are added, e.g., code rate ⅔ with BPSK, code rate ¾ with BPSK, and code rate ⅔ with QPSK to the existing MCS in the 802.11 standards. Using the combinations of repetition number, code rate, and modulation, a set of data rates is generated between 1.7-10 Mbps for ELR as follows. Since the delay spread is likely to be large for ELR channels, 0.8 microsecond cyclic prefix (CP) should not be used especially for the uplink. Instead, 1.6 or 3.2 microseconds of CP should be used. For example, 1.6 microseconds may be used for both downlink and uplink, and 3.2 microseconds may be used for multiuser uplink. In the Table 2, 1.6 microseconds of CP are used.

TABLE-US-00002 TABLE 2 MCSs from 52-tone RU.

Modulation	Code Rate	Repetition Number	Date Rate (Mb/s)
BPSK	1/2	4	1.7
BPSK	2/3	4	2.2
BPSK	3/4	4	2.5
QPSK	1/2	4	3.3
QPSK	2/3	4	4.4
QPSK	3/4	4	5.0
BPSK	1/2	2	3.3
BPSK	2/3	2	4.4
BPSK	3/4	2	5.0
QPSK	1/2	2	6.6
QPSK	2/3	2	8.8
QPSK	3/4	2	10.0

[0043] About 3-6 combinations may be selected with the target data rates, which have good performances in fading channels for BCC and LDPC codes, from the table above for ELR. It is likely that the lowest data rate of ELR may be about 1.5 Mbps instead of 1 Mbps. This lowest data rate excludes few options in the table. If code rate ¾ with BPSK is not added to the existing MCS, few more options are excluded from the table. If two options have about the same data rate, the option with lower code rate may be selected for ELR.

[0044] To improve the power efficiency, the middle 26-tone RU may be used at the cost of complexity. The middle 26-tone RU carries the QAM symbols, which are down sampled from Data 1 in FIG. **5**A and from both Data 1 and Data 2 in FIG. **5**B. This is illustrated, respectively. In FIG. **6**. The indexes of the data subcarriers of the sampled QAM symbols should be defined in the standard. For supporting block convolutional code (BCC), the codebits in the sampled QAM symbols should be distributed in the BCC codebit sequence (before interleaving) as evenly as possible.

Bit Repetitions within 242-tone RU

[0045] Because it is hard to utilize the middle 26-tone RU, 242-tone RU with codebit repetition can be considered for ELR. For reusing the existing 242-tone RU structure, the number of pilot subcarriers is still 8. The pilots may be power boosted for tracking the CFO in the low SNRs of ELR. The power boosting may be 3-6 dB. The number of data subcarriers is still 234. It is desired that the original codebits and their repetitions are all in the same OFDM symbol so that the log-likelihood ratio (LLR) combining of the codebits doesn't need to be across OFDM symbols for reducing latency and complexity. Note that the number 234 can be factored as 2×3×3×13. Therefore, the numbers repetitions can be 2, 3, 6, 9, 13, 18, and 26.

[0046] The target data rates of ELR range from 1 Mbps to 8 Mbps. To achieve the target data rates, three combinations of code rate and modulation are added, e.g., code rate ⅔ with BPSK, code rate ¾ with BPSK, and code rate ⅔ with QPSK to the existing MCS in the 802.11 standards. Using the combinations of repetition number, code rate, and modulation, a set of data rates between 0.9-10.8 Mbps for ELR are generated as follows. Since the delay spread is likely to be large for ELR channels, 0.8 microsecond cyclic prefix (CP) should not be used especially for the uplink. Instead, 1.6 or 3.2 microseconds of CP should be used. For example, 1.6 microseconds may be used for both downlink and uplink, and 3.2 microseconds may be used for multiuser uplink. In the Table 3, 1.6 microseconds of CP are used.

TABLE-US-00003 TABLE 3 MCSs from 242-tone RU.

Modulation	Code Rate	Repetition Number	Date Rate (Mb/s)
BPSK	1/2	9	0.9
BPSK	2/3	9	1.2
BPSK	1/2	6	1.4
BPSK	3/4	9	1.4
BPSK	2/3	6	1.8
BPSK	3/4	6	2.1
QPSK	1/2	9	1.8
QPSK	2/3	9	2.4
QPSK	1/2	6	2.8
QPSK	3/4	9	2.8
QPSK	2/3	6	3.6

QPSK 3/4 6 4.2 BPSK 1/2 3 2.7 BPSK 2/3 3 3.6 BPSK 3/4 3 4.2 QPSK 1/2 3 5.4 QPSK 2/3 3 7.2 QPSK 3/4 3 10.8

[0047] About 3-6 combinations may be picked with the target data rates, which have good performances in fading channels for BCC and LDPC codes, from the table above for ELR. It is likely that the lowest data rate of ELR may be about 1.5 Mbps instead of 1 Mbps. This lowest data rate excludes a few options in the table. If code rate ¾ with BPSK is not added to the existing MCS, few more options are excluded from the table. If two options have about the same data rate, the option with a lower code rate may be selected for ELR.

[0048] The codebit repetition can be done together with a block interleaver over the 242 data subcarriers for enhancing the diversity. An example is illustrated in FIG. **7**. The block interleaver reads bits in column by column and reads the bits out row by row as illustrated in the upper part of FIG. **7**. The each codebit may be first repeated by N times, where N is the number of repetitions, then the repeated codebits are concatenated, and sent to the interleaver. Alternatively, the data subcarriers are filled sequentially by the codebits from the lower frequency to the higher frequency as illustrated in the lower part of FIG. **7** (or from the higher frequency to the lower). The repetition number is 6 in FIG. **7**. For each OFDM symbol, 39 QAM symbols like BPSK or QPSK carry the codebits of the FEC encoder like BCC and LDPC. The 39 QAM symbols are sequentially filled in a segment of 39 adjacent data subcarriers and the process is repeated 6 times for filling up the 234 data subcarriers. The 8 pilot subcarriers are inserted in the 234 data subcarriers before transmission.

Modulation and Coding for ELR-SIG

[0049] For backward compatibility, it is likely that RU duplication like the 26-tone or 52-tone RU repetition aforementioned will be adopted by IEEE 802.11bn for modulating the data portion of the ELR PPDU. The modulation scheme for the PHY preamble is proposed, i.e., ELR-SIG field, in this subsection.

[0050] If the ELR-SIG field is in error, the whole ELR PPDU is lost. Therefore, the modulation and coding scheme (MCS) of ELR-SIG should be more robust than the data portion of the ELR PPDU. Although the data portion of ELR PPDU can use either BCC or more powerful LDPC, the coding scheme of ELR-SIG should be the BCC instead of the LDPC because the short length of ELR-SIG does not fill the shortest LDPC codeword well without severe shortening. Namely, it is still likely for the ELR-SIG to use BPSK, code rate ½, length 648 LDPC with shortening about half of the systematic bits, the chance is low. As a result, either the modulation order or the code rate needs to be lower for the ELR-SIG than the data portion. Or the number of repetitions of the ELR-SIG needs to be greater for the ELR-SIR than the data portion. For example, if both the ELR-SIG and the data portion use 52-tone RU with 4-time repetition, then the ELR-SIG needs codebit repetition in addition to the RU repetition for the additional reliability. For another example, if both the ELR-SIG and the data portion use 26-tone RU with repetition, then the repetition number of the ELR-SIG should be larger than that of the data portion, e.g., 9 vs 3. For a third example, if both the ELR-SIG and the data portion use 242-tone RU with codebit repetition, then the repletion number of ELR-SIG can be greater than that of the data portion, e.g., 6 vs 3, for enhanced reliability. For a fourth example, the modulation order and/or the code rate of the ELR-SIG can be lower than that of the data portion, e.g., BPSK vs QPSK and code rate ½ vs ⅔, for enhanced reliability.

[0051] For low complexity, it is likely that both the ELR-SIG and the data portion use the same RU repetition as 52-tone RU repetition. For additional reliability, the ELR-SIG may use 2-time codebit repetition in addition to the RU repetition. For the example of 52-tone RU repetition, the ELR-SIG uses BPSK with code rate 1/2 BCC for each 52-tone RU and the four 52-tone RUs within the same OFDM symbol carry the same data and the same BPSK data symbols. For additional reliability, 2-time codebit repetition is applied to each 52-tone RU in the same manner. Namely, only 24 codebits instead of 48 codebits of BCC are carried by each 52-tone RU. For each 52-tone RU, the 24 data subcarriers at the lower frequency carry the same codebits as the higher frequency. This is like the DCM of 802.11ax. Namely, ELR-SIG uses 52-tone RU with DCM, BPSK, BCC code rate ½, and

4-time RU repetition, and 20 MHz bandwidth. Namely, for each OFDM symbol, 24 codebits are repeated 8 times in frequency domain. It is known that the frequency domain repetition increases the PAPR. The PAPR reduction is described in the next subsection. As an alternative of the codebit repetition, time repetition may be used. For example, each OFDM symbol of the ELR-SIG may be sent twice in a row to boost the received SNR by 3 dB. The downside of this scheme is that the frequency diversity is slightly worse than the codebit repetition and the receiver needs to buffer two OFDM symbols before combining them.

[0052] The RU repetition, codebit repetition, and time repetition can be used jointly or individually for enhancing the reliability of the ELR-SIG.

PAPR Reduction

[0053] To reduce the PAPR, different phase rotations can be applied to the frequency-domain repetitions, respectively like DCM and the non-HT duplicate transmission in 802.11ax. For example, the 8 phases of the phase rotation sequence, which are applied to each 20 MHz subchannel of the 160 MHz non-HT duplicate PPDU, may be reused for the ELR-SIG (or the data portion) with 4-time 52-tone RU repetition and 2-time codebit repetition. For another example, the phase rotation sequence for each 20 MHz subchannel of the 80 MHz non-HT duplicate PPDU may be reused for the ELR-SIG with 4-time 52-tone RU repetition without codebit repetition. For a third example, the phase rotation sequence for each 20 MHz subchannel of the 80 MHz non-HT duplicate PPDU may be used for the ELR data portion with 4-time 52-tone RU repetition. To further reduce the PAPR, different cyclic shift diversity (CSD) valves can be applied to the frequency-domain repetitions, respectively. The phase rotation and CSD can be applied jointly or individually. Other ideas to reduce the PAPR include 1) using different masking sequences on different RUs to randomize the repeated data symbols in different ways, and 2) using different pilot sequences for different RUs to reduce the level of repetition in frequency domain.

[0054] It is understood that the above descriptions are for the purposes of illustration and are not meant to be limiting.

[0055] FIG. **8** illustrates a flow diagram of illustrative process **800** for an enhanced data modulation system, in accordance with one or more example embodiments of the present disclosure.

[0056] At block **802**, a device (e.g., the user device(s) **120** and/or the AP **102** of FIG. **1** and/or the enhanced data modulation device **1019** of FIG. **10**) may receive data to be transmitted over a wireless communication channel using a 20 MHz bandwidth and a plurality of 26-tone resource units (RUs).

[0057] At block **804**, the device may determine a repetition type for the plurality of 26-tone RUs based on a transmission configuration, wherein the repetition type is selected from a first repetition and a second repetition.

[0058] At block **806**, the device may map the data to the plurality of 26-tone RUs based on the repetition type, wherein the first repetition maps different data sets to subsets of the plurality of 26-tone RUs, each repeated three times, and the second repetition maps identical data across the plurality of 26-tone RUs.

[0059] At block **808**, the device may transmit the data over the wireless communication channel using the plurality of 26-tone RUs.

[0060] In one or more embodiments, the device may implement the first repetition by mapping three distinct data sets to a group of nine 26-tone RUs, with each data set repeated on three RUs, and may implement the second repetition by mapping a single data set identically across the nine 26-tone RUs. The device may reuse predefined RU configurations comprising 26-tone, 52-tone, 106-tone, and 242-tone RUs to maintain backward compatibility. The device may determine the repetition type based on a signal quality metric or a target range performance. The device may perform round-robin placement of repeated data using the plurality of 26-tone RUs to enhance frequency diversity gain. The device may assign unique indices to each 26-tone RU of the plurality of 26-tone RUs when mapping the data. The device may encode the data prior to mapping using a forward error correction code.

[0061] It is understood that the above descriptions are for the purposes of illustration and are not meant to be limiting.

[0062] FIG. **9** shows a functional diagram of an exemplary communication station **900**, in accordance with one or more example embodiments of the present disclosure. In one embodiment, FIG. **9** illustrates a functional block diagram of a communication station that may be suitable for use as an AP **102** (FIG. **1**) or a user device **120** (FIG. **1**) in accordance with some embodiments. The communication station **900** may also be suitable for use as a handheld device, a mobile device, a cellular telephone, a smartphone, a tablet, a netbook, a wireless terminal, a laptop computer, a wearable computer device, a femtocell, a high data rate (HDR) subscriber station, an access point, an access terminal, or other personal communication system (PCS) device.

[0063] The communication station **900** may include communications circuitry **902** and a transceiver **910** for transmitting and receiving signals to and from other communication stations using one or more antennas **901**. The communications circuitry **902** may include circuitry that can operate the physical layer (PHY) communications and/or medium access control (MAC) communications for controlling access to the wireless medium, and/or any other communications layers for transmitting and receiving signals. The communication station **900** may also include processing circuitry **906** and memory **908** arranged to perform the operations described herein. In some embodiments, the communications circuitry **902** and the processing circuitry **906** may be configured to perform operations detailed in the above figures, diagrams, and flows.

[0064] In accordance with some embodiments, the communications circuitry **902** may be arranged to contend for a wireless medium and configure frames or packets for communicating over the wireless medium. The communications circuitry **902** may be arranged to transmit and receive signals. The communications circuitry **902** may also include circuitry for modulation/demodulation, upconversion/downconversion, filtering, amplification, etc. In some embodiments, the processing circuitry **906** of the communication station **900** may include one or more processors. In other embodiments, two or more antennas **901** may be coupled to the communications circuitry **902** arranged for sending and receiving signals. The memory **908** may store information for configuring the processing circuitry **906** to perform operations for configuring and transmitting message frames and performing the various operations described herein. The memory **908** may include any type of memory, including non-transitory memory, for storing information in a form readable by a machine (e.g., a computer). For example, the memory **908** may include a computer-readable storage device, read-only memory (ROM), random-access memory (RAM), magnetic disk storage media, optical storage media, flash-memory devices and other storage devices and media.

[0065] In some embodiments, the communication station **900** may be part of a portable wireless communication device, such as a personal digital assistant (PDA), a laptop or portable computer with wireless communication capability, a web tablet, a wireless telephone, a smartphone, a wireless headset, a pager, an instant messaging device, a digital camera, an access point, a television, a medical device (e.g., a heart rate monitor, a blood pressure monitor, etc.), a wearable computer device, or another device that may receive and/or transmit information wirelessly.

[0066] In some embodiments, the communication station **900** may include one or more antennas **901**. The antennas **901** may include one or more directional or omnidirectional antennas, including, for example, dipole antennas, monopole antennas, patch antennas, loop antennas, microstrip antennas, or other types of antennas suitable for transmission of RF signals. In some embodiments, instead of two or more antennas, a single antenna with multiple apertures may be used. In these embodiments, each aperture may be considered a separate antenna. In some multiple-input multiple-output (MIMO) embodiments, the antennas may be effectively separated for spatial diversity and the different channel characteristics that may result between each of the antennas and the antennas of a transmitting station.

[0067] In some embodiments, the communication station **900** may include one or more of a keyboard, a display, a non-volatile memory port, multiple antennas, a graphics processor, an application processor, speakers, and other mobile device elements. The display may be an LCD screen including a touch screen.

[0068] Although the communication station **900** is illustrated as having several separate functional elements, two or more of the functional elements may be combined and may be implemented by combinations of software-configured elements, such as processing elements including digital signal processors (DSPs), and/or other hardware elements. For example, some elements may include one or more microprocessors, DSPs, field-programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), radio-frequency integrated circuits (RFICs) and combinations of various hardware and logic circuitry for performing at least the functions described herein. In some embodiments, the functional elements of the communication station **900** may refer to one or more processes operating on one or more processing elements.

[0069] Certain embodiments may be implemented in one or a combination of hardware, firmware, and software. Other embodiments may also be implemented as instructions stored on a computer-readable storage device, which may be read and executed by at least one processor to perform the operations described herein. A computer-readable storage device may include any non-transitory memory mechanism for storing information in a form readable by a machine (e.g., a computer). For example, a computer-readable storage device may include read-only memory (ROM), random-access memory (RAM), magnetic disk storage media, optical storage media, flash-memory devices, and other storage devices and media. In some embodiments, the communication station **900** may include one or more processors and may be configured with instructions stored on a computer-readable storage device.

[0070] FIG. **10** illustrates a block diagram of an example of a machine **1000** or system upon which any one or more of the techniques (e.g., methodologies) discussed herein may be performed. In other embodiments, the machine **1000** may operate as a standalone device or may be connected (e.g., networked) to other machines. In a networked deployment, the machine **1000** may operate in the capacity of a server machine, a client machine, or both in server-client network environments. In an example, the machine **1000** may act as a peer machine in peer-to-peer (P2P) (or other distributed) network environments. The machine **1000** may be a personal computer (PC), a tablet PC, a set-top box (STB), a personal digital assistant (PDA), a mobile telephone, a wearable computer device, a web appliance, a network router, a switch or bridge, or any machine capable of executing instructions (sequential or otherwise) that specify actions to be taken by that machine, such as a base station. Further, while only a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein, such as cloud computing, software as a service (SaaS), or other computer cluster configurations.

[0071] Examples, as described herein, may include or may operate on logic or a number of components, modules, or mechanisms. Modules are tangible entities (e.g., hardware) capable of performing specified operations when operating. A module includes hardware. In an example, the hardware may be specifically configured to carry out a specific operation (e.g., hardwired). In another example, the hardware may include configurable execution units (e.g., transistors, circuits, etc.) and a computer readable medium containing instructions where the instructions configure the execution units to carry out a specific operation when in operation. The configuring may occur under the direction of the executions units or a loading mechanism. Accordingly, the execution units are communicatively coupled to the computer-readable medium when the device is operating. In this example, the execution units may be a member of more than one module. For example, under operation, the execution units may be configured by a first set of instructions to implement a first module at one point in time and reconfigured by a second set of instructions to implement a second module at a second point in time.

[0072] The machine (e.g., computer system) **1000** may include a hardware processor **1002** (e.g., a central processing unit (CPU), a graphics processing unit (GPU), a hardware processor core, or any combination thereof), a main memory **1004** and a static memory **1006**, some or all of which may communicate with each other via an interlink (e.g., bus) **1008**. The machine **1000** may further include a power management device **1032**, a graphics display device **1010**, an alphanumeric input device **1012** (e.g., a keyboard), and a user interface (UI) navigation device **1014** (e.g., a mouse). In an example, the graphics display device **1010**, alphanumeric input device **1012**, and UI navigation device **1014** may be a touch screen display. The machine **1000** may additionally include a storage device (i.e., drive unit) **1016**, a signal generation device **1018** (e.g., a speaker), an enhanced data modulation device **1019**, a network interface device/transceiver **1020** coupled to antenna(s) **1030**, and one or more sensors **1028**, such as a global positioning system (GPS) sensor, a compass, an accelerometer, or other sensor. The machine **1000** may include an output controller **1034**, such as a serial (e.g., universal serial bus (USB), parallel, or other wired or wireless (e.g., infrared (IR), near field communication (NFC), etc.) connection to communicate with or control one or more peripheral devices (e.g., a printer, a card reader, etc.)). The operations in accordance with one or more example embodiments of the present disclosure may be carried out by a baseband processor. The baseband processor may be configured to generate corresponding baseband signals. The baseband processor may further include physical layer (PHY) and medium access control layer (MAC) circuitry, and may further interface with the hardware processor **1002** for generation and processing of the baseband signals and for controlling operations of the main memory **1004**, the storage device **1016**, and/or the enhanced data modulation device **1019**. The baseband processor may be provided on a single radio card, a single chip, or an integrated circuit (IC).

[0073] The storage device **1016** may include a machine readable medium **1022** on which is stored one or more sets of data structures or instructions **1024** (e.g., software) embodying or utilized by any one or more of the techniques or functions described herein. The instructions **1024** may also reside, completely or at least partially, within the main memory **1004**, within the static memory **1006**, or within the hardware processor **1002** during execution thereof by the machine **1000**. In an example, one or any combination of the hardware processor **1002**, the main memory **1004**, the static memory **1006**, or the storage device **1016** may constitute machine-readable media.

[0074] The enhanced data modulation device **1019** may carry out or perform any of the operations and processes (e.g., process **800**) described and shown above.

[0075] It is understood that the above are only a subset of what the enhanced data modulation device **1019** may be configured to perform and that other functions included throughout this disclosure may also be performed by the enhanced data modulation device **1019**.

[0076] While the machine-readable medium **1022** is illustrated as a single medium, the term “machine-readable medium” may include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) configured to store the one or more instructions **1024**.

[0077] Various embodiments may be implemented fully or partially in software and/or firmware. This software and/or firmware may take the form of instructions contained in or on a non-transitory computer-readable storage medium. Those instructions may then be read and executed by one or more processors to enable performance of the operations described herein. The instructions may be in any suitable form, such as but not limited to source code, compiled code, interpreted code, executable code, static code, dynamic code, and the like. Such a computer-readable medium may include any tangible non-transitory medium for storing information in a form readable by one or more computers, such as but not limited to read only memory (ROM); random access memory (RAM); magnetic disk storage media; optical storage media; a flash memory, etc.

[0078] The term “machine-readable medium” may include any medium that is capable of storing, encoding, or carrying instructions for execution by the machine **1000** and that cause the machine

1000 to perform any one or more of the techniques of the present disclosure, or that is capable of storing, encoding, or carrying data structures used by or associated with such instructions. Non-limiting machine-readable medium examples may include solid-state memories and optical and magnetic media. In an example, a massed machine-readable medium includes a machine-readable medium with a plurality of particles having resting mass. Specific examples of massed machine-readable media may include non-volatile memory, such as semiconductor memory devices (e.g., electrically programmable read-only memory (EPROM), or electrically erasable programmable read-only memory (EEPROM)) and flash memory devices; magnetic disks, such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks.

[0079] The instructions **1024** may further be transmitted or received over a communications network **1026** using a transmission medium via the network interface device/transceiver **1020** utilizing any one of a number of transfer protocols (e.g., frame relay, internet protocol (IP), transmission control protocol (TCP), user datagram protocol (UDP), hypertext transfer protocol (HTTP), etc.). Example communications networks may include a local area network (LAN), a wide area network (WAN), a packet data network (e.g., the Internet), mobile telephone networks (e.g., cellular networks), plain old telephone (POTS) networks, wireless data networks (e.g., Institute of Electrical and Electronics Engineers (IEEE) 802.11 family of standards known as Wi-Fi®, IEEE 802.16 family of standards known as WiMax®), IEEE 802.15.4 family of standards, and peer-to-peer (P2P) networks, among others. In an example, the network interface device/transceiver **1020** may include one or more physical jacks (e.g., Ethernet, coaxial, or phone jacks) or one or more antennas to connect to the communications network **1026**. In an example, the network interface device/transceiver **1020** may include a plurality of antennas to wirelessly communicate using at least one of single-input multiple-output (SIMO), multiple-input multiple-output (MIMO), or multiple-input single-output (MISO) techniques. The term “transmission medium” shall be taken to include any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the machine **1000** and includes digital or analog communications signals or other intangible media to facilitate communication of such software.

[0080] The operations and processes described and shown above may be carried out or performed in any suitable order as desired in various implementations. Additionally, in certain implementations, at least a portion of the operations may be carried out in parallel. Furthermore, in certain implementations, less than or more than the operations described may be performed.

[0081] FIG. **11** is a block diagram of a radio architecture **105**A, **105**B in accordance with some embodiments that may be implemented in any one of the example APs **102** and/or the example STAs **120** of FIG. **1**. Radio architecture **105**A, **105**B may include radio front-end module (FEM) circuitry **1104***a-b*, radio IC circuitry **1106***a-b* and baseband processing circuitry **1108***a-b*. Radio architecture **105**A, **105**B as shown includes both Wireless Local Area Network (WLAN) functionality and Bluetooth (BT) functionality although embodiments are not so limited. In this disclosure, “WLAN” and “Wi-Fi” are used interchangeably.

[0082] FEM circuitry **1104***a-b* may include a WLAN or Wi-Fi FEM circuitry **1104***a* and a Bluetooth (BT) FEM circuitry **1104***b*. The WLAN FEM circuitry **1104***a* may include a receive signal path comprising circuitry configured to operate on WLAN RF signals received from one or more antennas **1101**, to amplify the received signals and to provide the amplified versions of the received signals to the WLAN radio IC circuitry **1106***a* for further processing. The BT FEM circuitry **1104***b* may include a receive signal path which may include circuitry configured to operate on BT RF signals received from one or more antennas **1101**, to amplify the received signals and to provide the amplified versions of the received signals to the BT radio IC circuitry **1106***b* for further processing. FEM circuitry **1104***a* may also include a transmit signal path which may include circuitry configured to amplify WLAN signals provided by the radio IC circuitry **1106***a* for wireless transmission by one or more of the antennas **1101**. In addition, FEM circuitry **1104***b* may also include a transmit signal path which may include circuitry configured to amplify BT signals provided by the radio IC circuitry **1106***b* for wireless transmission by the one or more antennas. In the embodiment of FIG. **11**, although FEM **1104***a* and FEM **1104***b* are shown as being distinct from one another, embodiments are not so limited, and include within their scope the use of an FEM (not shown) that includes a transmit path and/or a receive path for both WLAN and BT signals, or the use of one or more FEM circuitries where at least some of the FEM circuitries share transmit and/or receive signal paths for both WLAN and BT signals.

[0083] Radio IC circuitry **1106***a-b* as shown may include WLAN radio IC circuitry **1106***a* and BT radio IC circuitry **1106***b*. The WLAN radio IC circuitry **1106***a* may include a receive signal path which may include circuitry to down-convert WLAN RF signals received from the FEM circuitry **1104***a* and provide baseband signals to WLAN baseband processing circuitry **1108***a*. BT radio IC circuitry **1106***b* may in turn include a receive signal path which may include circuitry to down-convert BT RF signals received from the FEM circuitry **1104***b* and provide baseband signals to BT baseband processing circuitry **1108***b*. WLAN radio IC circuitry **1106***a* may also include a transmit signal path which may include circuitry to up-convert WLAN baseband signals provided by the WLAN baseband processing circuitry **1108***a* and provide WLAN RF output signals to the FEM circuitry **1104***a* for subsequent wireless transmission by the one or more antennas **1101**. BT radio IC circuitry **1106***b* may also include a transmit signal path which may include circuitry to up-convert BT baseband signals provided by the BT baseband processing circuitry **1108***b* and provide BT RF output signals to the FEM circuitry **1104***b* for subsequent wireless transmission by the one or more antennas **1101**. In the embodiment of FIG. **11**, although radio IC circuitries **1106***a* and **1106***b* are shown as being distinct from one another, embodiments are not so limited, and include within their scope the use of a radio IC circuitry (not shown) that includes a transmit signal path and/or a receive signal path for both WLAN and BT signals, or the use of one or more radio IC circuitries where at least some of the radio IC circuitries share transmit and/or receive signal paths for both WLAN and BT signals.

[0084] Baseband processing circuity **1108***a-b* may include a WLAN baseband processing circuitry **1108***a* and a BT baseband processing circuitry **1108***b*. The WLAN baseband processing circuitry **1108***a* may include a memory, such as, for example, a set of RAM arrays in a Fast Fourier Transform or Inverse Fast Fourier Transform block (not shown) of the WLAN baseband processing circuitry **1108***a*. Each of the WLAN baseband circuitry **1108***a* and the BT baseband circuitry **1108***b* may further include one or more processors and control logic to process the signals received from the corresponding WLAN or BT receive signal path of the radio IC circuitry **1106***a-b*, and to also generate corresponding WLAN or BT baseband signals for the transmit signal path of the radio IC circuitry **1106***a-b*. Each of the baseband processing circuitries **1108***a* and **1108***b* may further include physical layer (PHY) and medium access control layer (MAC) circuitry, and may further interface with a device for generation and processing of the baseband signals and for controlling operations of the radio IC circuitry **1106***a-b*.

[0085] Referring still to FIG. **11**, according to the shown embodiment, WLAN-BT coexistence circuitry **1113** may include logic providing an interface between the WLAN baseband circuitry **1108***a* and the BT baseband circuitry **1108***b* to enable use cases requiring WLAN and BT coexistence. In addition, a switch **1103** may be provided between the WLAN FEM circuitry **1104***a* and the BT FEM circuitry **1104***b* to allow switching between the WLAN and BT radios according to application needs. In addition, although the antennas **1101** are depicted as being respectively connected to the WLAN FEM circuitry **1104***a* and the BT FEM circuitry **1104***b*, embodiments include within their scope the sharing of one or more antennas as between the WLAN and BT FEMs, or the provision of more than one antenna connected to each of FEM **1104***a* or **1104***b*.

[0086] In some embodiments, the front-end module circuitry **1104***a-b*, the radio IC circuitry **1106***a-b*, and baseband processing circuitry **1108***a-b* may be provided on a single radio card, such as wireless radio card **1102**. In some other embodiments, the one or more antennas **1101**, the FEM circuitry **1104***a-b* and the radio IC circuitry **1106***a-b* may be provided on a single radio card. In some other embodiments, the radio IC circuitry **1106***a-b* and the baseband processing circuitry **1108***a-b* may be provided on a single chip or integrated circuit (IC), such as IC **1112**.

[0087] In some embodiments, the wireless radio card **1102** may include a WLAN radio card and may be configured for Wi-Fi communications, although the scope of the embodiments is not limited in this respect. In some of these embodiments, the radio architecture **105**A, **105**B may be configured to receive and transmit orthogonal frequency division multiplexed (OFDM) or orthogonal frequency division multiple access (OFDMA) communication signals over a multicarrier communication channel. The OFDM or OFDMA signals may comprise a plurality of orthogonal subcarriers.

[0088] In some of these multicarrier embodiments, radio architecture **105**A, **105**B may be part of a Wi-Fi communication station (STA) such as a wireless access point (AP), a base station or a mobile device including a Wi-Fi device. In some of these embodiments, radio architecture **105**A, **105**B may be configured to transmit and receive signals in accordance with specific communication standards and/or protocols, such as any of the Institute of Electrical and Electronics Engineers (IEEE) standards including, 802.11n-2009, IEEE 802.11-2012, IEEE 802.11-2016, 802.11n-2009, 802.11ac, 802.11ah, 802.11ad, 802.11ay and/or 802.11ax standards and/or proposed specifications for WLANs, although the scope of embodiments is not limited in this respect. Radio architecture **105**A, **105**B may also be suitable to transmit and/or receive communications in accordance with other techniques and standards.

[0089] In some embodiments, the radio architecture **105**A, **105**B may be configured for high-efficiency Wi-Fi (HEW) communications in accordance with the IEEE 802.11ax standard. In these embodiments, the radio architecture **105**A, **105**B may be configured to communicate in accordance with an OFDMA technique, although the scope of the embodiments is not limited in this respect.

[0090] In some other embodiments, the radio architecture **105**A, **105**B may be configured to transmit and receive signals transmitted using one or more other modulation techniques such as spread spectrum modulation (e.g., direct sequence code division multiple access (DS-CDMA) and/or frequency hopping code division multiple access (FH-CDMA)), time-division multiplexing (TDM) modulation, and/or frequency-division multiplexing (FDM) modulation, although the scope of the embodiments is not limited in this respect.

[0091] In some embodiments, as further shown in FIG. **6**, the BT baseband circuitry **1108***b* may be compliant with a Bluetooth (BT) connectivity standard such as Bluetooth, Bluetooth 8.0 or Bluetooth 6.0, or any other iteration of the Bluetooth Standard.

[0092] In some embodiments, the radio architecture **105**A, **105**B may include other radio cards, such as a cellular radio card configured for cellular (e.g., 5GPP such as LTE, LTE-Advanced or 7G communications).

[0093] In some IEEE 802.11 embodiments, the radio architecture **105**A, **105**B may be configured for communication over various channel bandwidths including bandwidths having center frequencies of about 900 MHz, 2.4 GHz, 5 GHZ, and bandwidths of about 2 MHz, 4 MHZ, 5 MHz, 5.5 MHz, 6 MHz, 8 MHz, 10 MHz, 20 MHz, 40 MHz, 80 MHZ (with contiguous bandwidths) or 80+80 MHz (160 MHz) (with non-contiguous bandwidths). In some embodiments, a 920 MHz channel bandwidth may be used. The scope of the embodiments is not limited with respect to the above center frequencies however.

[0094] FIG. **12** illustrates WLAN FEM circuitry **1104***a* in accordance with some embodiments. Although the example of FIG. **12** is described in conjunction with the WLAN FEM circuitry **1104***a*, the example of FIG. **12** may be described in conjunction with the example BT FEM circuitry **1104***b* (FIG. **11**), although other circuitry configurations may also be suitable.

[0095] In some embodiments, the FEM circuitry **1104***a* may include a TX/RX switch **1202** to switch between transmit mode and receive mode operation. The FEM circuitry **1104***a* may include a receive signal path and a transmit signal path. The receive signal path of the FEM circuitry **1104***a* may include a low-noise amplifier (LNA) **1206** to amplify received RF signals **1203** and provide the amplified received RF signals **1207** as an output (e.g., to the radio IC circuitry **1106***a-b* (FIG. **11**)). The transmit signal path of the circuitry **1104***a* may include a power amplifier (PA) to amplify input RF signals **1209** (e.g., provided by the radio IC circuitry **1106***a-b*), and one or more filters **1212**, such as band-pass filters (BPFs), low-pass filters (LPFs) or other types of filters, to generate RF signals **1215** for subsequent transmission (e.g., by one or more of the antennas **1101** (FIG. **11**)) via an example duplexer **1214**.

[0096] In some dual-mode embodiments for Wi-Fi communication, the FEM circuitry **1104***a* may be configured to operate in either the 2.4 GHz frequency spectrum or the 5 GHz frequency spectrum. In these embodiments, the receive signal path of the FEM circuitry **1104***a* may include a receive signal path duplexer **1204** to separate the signals from each spectrum as well as provide a separate LNA **1206** for each spectrum as shown. In these embodiments, the transmit signal path of the FEM circuitry **1104***a* may also include a power amplifier **1210** and a filter **1212**, such as a BPF, an LPF or another type of filter for each frequency spectrum and a transmit signal path duplexer **1204** to provide the signals of one of the different spectrums onto a single transmit path for subsequent transmission by the one or more of the antennas **1101** (FIG. **11**). In some embodiments, BT communications may utilize the 2.4 GHz signal paths and may utilize the same FEM circuitry **1104***a* as the one used for WLAN communications.

[0097] FIG. **13** illustrates radio IC circuitry **1106***a* in accordance with some embodiments. The radio IC circuitry **1106***a* is one example of circuitry that may be suitable for use as the WLAN or BT radio IC circuitry **1106***a*/**1106***b* (FIG. **11**), although other circuitry configurations may also be suitable. Alternatively, the example of FIG. **13** may be described in conjunction with the example BT radio IC circuitry **1106***b*.

[0098] In some embodiments, the radio IC circuitry **1106***a* may include a receive signal path and a transmit signal path. The receive signal path of the radio IC circuitry **1106***a* may include at least mixer circuitry **1302**, such as, for example, down-conversion mixer circuitry, amplifier circuitry **1306** and filter circuitry **1308**. The transmit signal path of the radio IC circuitry **1106***a* may include at least filter circuitry **1312** and mixer circuitry **1314**, such as, for example, up-conversion mixer circuitry. Radio IC circuitry **1106***a* may also include synthesizer circuitry **1304** for synthesizing a frequency **1305** for use by the mixer circuitry **1302** and the mixer circuitry **1314**. The mixer circuitry **1302** and/or **1314** may each, according to some embodiments, be configured to provide direct conversion functionality. The latter type of circuitry presents a much simpler architecture as compared with standard super-heterodyne mixer circuitries, and any flicker noise brought about by the same may be alleviated for example through the use of OFDM modulation. FIG. **13** illustrates only a simplified version of a radio IC circuitry, and may include, although not shown, embodiments where each of the depicted circuitries may include more than one component. For instance, mixer circuitry **1314** may each include one or more mixers, and filter circuitries **1308** and/or **1312** may each include one or more filters, such as one or more BPFs and/or LPFs according to application needs. For example, when mixer circuitries are of the direct-conversion type, they may each include two or more mixers.

[0099] In some embodiments, mixer circuitry **1302** may be configured to down-convert RF signals **1207** received from the FEM circuitry **1104***a-b* (FIG. **11**) based on the synthesized frequency **1305** provided by synthesizer circuitry **1304**. The amplifier circuitry **1306** may be configured to amplify the down-converted signals and the filter circuitry **1308** may include an LPF configured to remove unwanted signals from the down-converted signals to generate output baseband signals **1307**. Output baseband signals **1307** may be provided to the baseband processing circuitry **1108***a-b* (FIG. **11**) for further processing. In some embodiments, the output baseband signals **1307** may be zero-frequency baseband signals, although this is not a requirement. In some embodiments, mixer circuitry **1302** may comprise passive mixers, although the scope of the embodiments is not limited in this respect.

[0100] In some embodiments, the mixer circuitry **1314** may be configured to up-convert input baseband signals **1311** based on the synthesized frequency **1305** provided by the synthesizer circuitry **1304** to generate RF output signals **1209** for the FEM circuitry **1104***a-b*. The baseband signals **1311** may be provided by the baseband processing circuitry **1108***a-b* and may be filtered by filter circuitry **1312**. The filter circuitry **1312** may include an LPF or a BPF, although the scope of the embodiments is not limited in this respect.

[0101] In some embodiments, the mixer circuitry **1302** and the mixer circuitry **1314** may each include two or more mixers and may be arranged for quadrature down-conversion and/or up-conversion respectively with the help of synthesizer **1304**. In some embodiments, the mixer circuitry **1302** and the mixer circuitry **1314** may each include two or more mixers each configured for image rejection (e.g., Hartley image rejection). In some embodiments, the mixer circuitry **1302** and the mixer circuitry **1314** may be arranged for direct down-conversion and/or direct up-conversion, respectively. In some embodiments, the mixer circuitry **1302** and the mixer circuitry **1314** may be configured for super-heterodyne operation, although this is not a requirement.

[0102] Mixer circuitry **1302** may comprise, according to one embodiment: quadrature passive mixers (e.g., for the in-phase (I) and quadrature phase (Q) paths). In such an embodiment, RF input signal **1207** from FIG. **13** may be down-converted to provide I and Q baseband output signals to be sent to the baseband processor.

[0103] Quadrature passive mixers may be driven by zero and ninety-degree time-varying LO switching signals provided by a quadrature circuitry which may be configured to receive a LO frequency (fLO) from a local oscillator or a synthesizer, such as LO frequency **1305** of synthesizer **1304** (FIG. **13**). In some embodiments, the LO frequency may be the carrier frequency, while in other embodiments, the LO frequency may be a fraction of the carrier frequency (e.g., one-half the carrier frequency, one-third the carrier frequency). In some embodiments, the zero and ninety-degree time-varying switching signals may be generated by the synthesizer, although the scope of the embodiments is not limited in this respect.

[0104] In some embodiments, the LO signals may differ in duty cycle (the percentage of one period in which the LO signal is high) and/or offset (the difference between start points of the period). In some embodiments, the LO signals may have an 85% duty cycle and an 80% offset. In some embodiments, each branch of the mixer circuitry (e.g., the in-phase (I) and quadrature phase (Q) path) may operate at an 80% duty cycle, which may result in a significant reduction is power consumption.

[0105] The RF input signal **1207** (FIG. **12**) may comprise a balanced signal, although the scope of the embodiments is not limited in this respect. The I and Q baseband output signals may be provided to low-noise amplifier, such as amplifier circuitry **1306** (FIG. **13**) or to filter circuitry **1308** (FIG. **13**).

[0106] In some embodiments, the output baseband signals **1307** and the input baseband signals **1311** may be analog baseband signals, although the scope of the embodiments is not limited in this respect. In some alternate embodiments, the output baseband signals **1307** and the input baseband signals **1311** may be digital baseband signals. In these alternate embodiments, the radio IC circuitry may include analog-to-digital converter (ADC) and digital-to-analog converter (DAC) circuitry.

[0107] In some dual-mode embodiments, a separate radio IC circuitry may be provided for processing signals for each spectrum, or for other spectrums not mentioned here, although the scope of the embodiments is not limited in this respect.

[0108] In some embodiments, the synthesizer circuitry **1304** may be a fractional-N synthesizer or a fractional N/N+1 synthesizer, although the scope of the embodiments is not limited in this respect as other types of frequency synthesizers may be suitable. For example, synthesizer circuitry **1304** may be a delta-sigma synthesizer, a frequency multiplier, or a synthesizer comprising a phase-locked loop with a frequency divider. According to some embodiments, the synthesizer circuitry **1304** may include digital synthesizer circuitry. An advantage of using a digital synthesizer circuitry is that, although it may still include some analog components, its footprint may be scaled down

much more than the footprint of an analog synthesizer circuitry. In some embodiments, frequency input into synthesizer circuitry **1304** may be provided by a voltage controlled oscillator (VCO), although that is not a requirement. A divider control input may further be provided by either the baseband processing circuitry **1108***a-b* (FIG. **11**) depending on the desired output frequency **1305**. In some embodiments, a divider control input (e.g., N) may be determined from a look-up table (e.g., within a Wi-Fi card) based on a channel number and a channel center frequency as determined or indicated by the example application processor **1110**. The application processor **1110** may include, or otherwise be connected to, one of the example secure signal converter **101** or the example received signal converter **103** (e.g., depending on which device the example radio architecture is implemented in).
[0109] In some embodiments, synthesizer circuitry **1304** may be configured to generate a carrier frequency as the output frequency **1305**, while in other embodiments, the output frequency **1305** may be a fraction of the carrier frequency (e.g., one-half the carrier frequency, one-third the carrier frequency). In some embodiments, the output frequency **1305** may be a LO frequency (fLO).
[0110] FIG. Q**14** illustrates a functional block diagram of baseband processing circuitry **1108***a* in accordance with some embodiments. The baseband processing circuitry **1108***a* is one example of circuitry that may be suitable for use as the baseband processing circuitry **1108***a* (FIG. **11**), although other circuitry configurations may also be suitable. Alternatively, the example of FIG. **13** may be used to implement the example BT baseband processing circuitry **1108***b* of FIG. **11**.
[0111] The baseband processing circuitry **1108***a* may include a receive baseband processor (RX BBP) Q**1402** for processing receive baseband signals **1309** provided by the radio IC circuitry **1106***a-b* (FIG. **11**) and a transmit baseband processor (TX BBP) Q**1404** for generating transmit baseband signals **1311** for the radio IC circuitry **1106***a-b*. The baseband processing circuitry **1108***a* may also include control logic Q**1406** for coordinating the operations of the baseband processing circuitry **1108***a*.
[0112] In some embodiments (e.g., when analog baseband signals are exchanged between the baseband processing circuitry **1108***a-b* and the radio IC circuitry **1106***a-b*), the baseband processing circuitry **1108***a* may include ADC Q**1410** to convert analog baseband signals Q**1409** received from the radio IC circuitry **1106***a-b* to digital baseband signals for processing by the RX BBP Q**1402**. In these embodiments, the baseband processing circuitry **1108***a* may also include DAC Q**1412** to convert digital baseband signals from the TX BBP Q**1404** to analog baseband signals Q**1411**.
[0113] In some embodiments that communicate OFDM signals or OFDMA signals, such as through baseband processor **1108***a*, the transmit baseband processor Q**1404** may be configured to generate OFDM or OFDMA signals as appropriate for transmission by performing an inverse fast Fourier transform (IFFT). The receive baseband processor Q**1402** may be configured to process received OFDM signals or OFDMA signals by performing an FFT. In some embodiments, the receive baseband processor Q**1402** may be configured to detect the presence of an OFDM signal or OFDMA signal by performing an autocorrelation, to detect a preamble, such as a short preamble, and by performing a cross-correlation, to detect a long preamble. The preambles may be part of a predetermined frame structure for Wi-Fi communication.
[0114] Referring back to FIG. **11**, in some embodiments, the antennas **1101** (FIG. **11**) may each comprise one or more directional or omnidirectional antennas, including, for example, dipole antennas, monopole antennas, patch antennas, loop antennas, microstrip antennas or other types of antennas suitable for transmission of RF signals. In some multiple-input multiple-output (MIMO) embodiments, the antennas may be effectively separated to take advantage of spatial diversity and the different channel characteristics that may result. Antennas **1101** may each include a set of phased-array antennas, although embodiments are not so limited.
[0115] Although the radio architecture **105**A, **105**B is illustrated as having several separate functional elements, one or more of the functional elements may be combined and may be implemented by combinations of software-configured elements, such as processing elements

including digital signal processors (DSPs), and/or other hardware elements. For example, some elements may comprise one or more microprocessors, DSPs, field-programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), radio-frequency integrated circuits (RFICs) and combinations of various hardware and logic circuitry for performing at least the functions described herein. In some embodiments, the functional elements may refer to one or more processes operating on one or more processing elements.

[0116] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments. The terms “computing device,” “user device,” “communication station,” “station,” “handheld device,” “mobile device,” “wireless device” and “user equipment” (UE) as used herein refers to a wireless communication device such as a cellular telephone, a smartphone, a tablet, a netbook, a wireless terminal, a laptop computer, a femtocell, a high data rate (HDR) subscriber station, an access point, a printer, a point of sale device, an access terminal, or other personal communication system (PCS) device. The device may be either mobile or stationary.

[0117] As used within this document, the term “communicate” is intended to include transmitting, or receiving, or both transmitting and receiving. This may be particularly useful in claims when describing the organization of data that is being transmitted by one device and received by another, but only the functionality of one of those devices is required to infringe the claim. Similarly, the bidirectional exchange of data between two devices (both devices transmit and receive during the exchange) may be described as “communicating,” when only the functionality of one of those devices is being claimed. The term “communicating” as used herein with respect to a wireless communication signal includes transmitting the wireless communication signal and/or receiving the wireless communication signal. For example, a wireless communication unit, which is capable of communicating a wireless communication signal, may include a wireless transmitter to transmit the wireless communication signal to at least one other wireless communication unit, and/or a wireless communication receiver to receive the wireless communication signal from at least one other wireless communication unit.

[0118] As used herein, unless otherwise specified, the use of the ordinal adjectives “first,” “second,” “third,” etc., to describe a common object, merely indicates that different instances of like objects are being referred to and are not intended to imply that the objects so described must be in a given sequence, either temporally, spatially, in ranking, or in any other manner.

[0119] The term “access point” (AP) as used herein may be a fixed station. An access point may also be referred to as an access node, a base station, an evolved node B (eNodeB), or some other similar terminology known in the art. An access terminal may also be called a mobile station, user equipment (UE), a wireless communication device, or some other similar terminology known in the art. Embodiments disclosed herein generally pertain to wireless networks. Some embodiments may relate to wireless networks that operate in accordance with one of the IEEE 802.11 standards.

[0120] Some embodiments may be used in conjunction with various devices and systems, for example, a personal computer (PC), a desktop computer, a mobile computer, a laptop computer, a notebook computer, a tablet computer, a server computer, a handheld computer, a handheld device, a personal digital assistant (PDA) device, a handheld PDA device, an on-board device, an off-board device, a hybrid device, a vehicular device, a non-vehicular device, a mobile or portable device, a consumer device, a non-mobile or non-portable device, a wireless communication station, a wireless communication device, a wireless access point (AP), a wired or wireless router, a wired or wireless modem, a video device, an audio device, an audio-video (A/V) device, a wired or wireless network, a wireless area network, a wireless video area network (WVAN), a local area network (LAN), a wireless LAN (WLAN), a personal area network (PAN), a wireless PAN (WPAN), and the like.

[0121] Some embodiments may be used in conjunction with one way and/or two-way radio communication systems, cellular radio-telephone communication systems, a mobile phone, a cellular telephone, a wireless telephone, a personal communication system (PCS) device, a PDA device which incorporates a wireless communication device, a mobile or portable global positioning system (GPS) device, a device which incorporates a GPS receiver or transceiver or chip, a device which incorporates an RFID element or chip, a multiple input multiple output (MIMO) transceiver or device, a single input multiple output (SIMO) transceiver or device, a multiple input single output (MISO) transceiver or device, a device having one or more internal antennas and/or external antennas, digital video broadcast (DVB) devices or systems, multi-standard radio devices or systems, a wired or wireless handheld device, e.g., a smartphone, a wireless application protocol (WAP) device, or the like.

[0122] Some embodiments may be used in conjunction with one or more types of wireless communication signals and/or systems following one or more wireless communication protocols, for example, radio frequency (RF), infrared (IR), frequency-division multiplexing (FDM), orthogonal FDM (OFDM), time-division multiplexing (TDM), time-division multiple access (TDMA), extended TDMA (E-TDMA), general packet radio service (GPRS), extended GPRS, code-division multiple access (CDMA), wideband CDMA (WCDMA), CDMA 2000, single-carrier CDMA, multi-carrier CDMA, multi-carrier modulation (MDM), discrete multi-tone (DMT), Bluetooth®, global positioning system (GPS), Wi-Fi, Wi-Max, ZigBee, ultra-wideband (UWB), global system for mobile communications (GSM), 2G, 2.5G, 3G, 3.5G, 4G, fifth generation (5G) mobile networks, 3GPP, long term evolution (LTE), LTE advanced, enhanced data rates for GSM Evolution (EDGE), or the like. Other embodiments may be used in various other devices, systems, and/or networks.

[0123] The following examples pertain to further embodiments.

[0124] Example 1 may include a device comprising processing circuitry coupled to storage, the processing circuitry configured to: receive data to be transmitted over a wireless communication channel using a 20 MHz bandwidth and a plurality of 26-tone resource units (RUS); determine a repetition type for the plurality of 26-tone RUs based on a transmission configuration, wherein the repetition type may be selected from a first repetition and a second repetition; map the data to the plurality of 26-tone RUs based on the repetition type, wherein the first repetition maps different data sets to subsets of the plurality of 26-tone RUs, each repeated three times, and the second repetition maps identical data across the plurality of 26-tone RUs; and transmit the data over the wireless communication channel using the plurality of 26-tone RUs.

[0125] Example 2 may include the device of example 1 and/or some other example(s) herein, wherein the first repetition comprises mapping three distinct data sets to a group of nine 26-tone RUs, with each data set repeated on three RUs, and the second repetition comprises mapping a single data set identically across the nine 26-tone RUs.

[0126] Example 3 may include the device of example 1 and/or some other example(s) herein, wherein the processing circuitry may be further configured to reuse predefined RU configurations comprising 26-tone, 52-tone, 106-tone, and 242-tone RUs to maintain backward compatibility.

[0127] Example 4 may include the device of example 1 and/or some other example(s) herein, wherein the repetition type may be determined based on a signal quality metric or a target range performance.

[0128] Example 5 may include the device of example 1 and/or some other example(s) herein, wherein the processing circuitry may be further configured to perform round-robin placement of repeated data using the plurality of 26-tone RUs to enhance frequency diversity gain.

[0129] Example 6 may include the device of example 1 and/or some other example(s) herein, wherein mapping the data may include assigning unique indices to each 26-tone RU of the plurality of 26-tone RUs.

[0130] Example 7 may include the device of example 1 and/or some other example(s) herein, wherein the processing circuitry may be further configured to encode the data prior to mapping using a forward error correction code.

[0131] Example 8 may include the device of example 1 and/or some other example(s) herein, further comprising a transceiver configured to transmit and receive wireless signals.

[0132] Example 9 may include the device of example 8 and/or some other example(s) herein, further comprising an antenna coupled to the transceiver to cause to send the data.

[0133] Example 10 may include a non-transitory computer-readable medium storing computer-executable instructions which when executed by one or more processors result in performing operations comprising: receiving data to be transmitted over a wireless communication channel using a 20 MHz bandwidth and a plurality of 26-tone resource units (RUs); determining a repetition type for the plurality of 26-tone RUs based on a transmission configuration, wherein the repetition type may be selected from a first repetition and a second repetition; mapping the data to the plurality of 26-tone RUs based on the repetition type, wherein the first repetition maps different data sets to subsets of the plurality of 26-tone RUs, each repeated three times, and the second repetition maps identical data across the plurality of 26-tone RUs; and transmitting the data over the wireless communication channel using the plurality of 26-tone RUs.

[0134] Example 11 may include the non-transitory computer-readable medium of example 10 and/or some other example(s) herein, wherein the first repetition comprises mapping three distinct data sets to a group of nine 26-tone RUs, with each data set repeated on three RUs, and the second repetition comprises mapping a single data set identically across the nine 26-tone RUs.

[0135] Example 12 may include the non-transitory computer-readable medium of example 10 and/or some other example(s) herein, wherein the operations further comprise reuse predefined RU configurations comprising 26-tone, 52-tone, 106-tone, and 242-tone RUs to maintain backward compatibility.

[0136] Example 13 may include the non-transitory computer-readable medium of example 10 and/or some other example(s) herein, wherein the repetition type may be determined based on a signal quality metric or a target range performance.

[0137] Example 14 may include the non-transitory computer-readable medium of example 10 and/or some other example(s) herein, wherein the operations further comprise performing round-robin placement of repeated data using the plurality of 26-tone RUs to enhance frequency diversity gain.

[0138] Example 15 may include the non-transitory computer-readable medium of example 10 and/or some other example(s) herein, wherein mapping the data may include assigning unique indices to each 26-tone RU of the plurality of 26-tone RUs.

[0139] Example 16 may include the non-transitory computer-readable medium of example 10 and/or some other example(s) herein, wherein the operations further comprise encoding the data prior to mapping using a forward error correction code.

[0140] Example 17 may include the non-transitory computer-readable medium of example 10 and/or some other example(s) herein, further comprising a transceiver configured to transmit and receive wireless signals.

[0141] Example 18 may include the non-transitory computer-readable medium of example 17 and/or some other example(s) herein, further comprising an antenna coupled to the transceiver to cause to send the data.

[0142] Example 19 may include a method comprising: receiving data to be transmitted over a wireless communication channel using a 20 MHz bandwidth and a plurality of 26-tone resource units (RUs); determining a repetition type for the plurality of 26-tone RUs based on a transmission configuration, wherein the repetition type may be selected from a first repetition and a second repetition; mapping the data to the plurality of 26-tone RUs based on the repetition type, wherein the first repetition maps different data sets to subsets of the plurality of 26-tone RUs, each repeated three times, and the second repetition maps identical data across the plurality of 26-tone RUs; and transmitting the data over the wireless communication channel using the plurality of 26-tone RUs.

[0143] Example 20 may include the method of example 19 and/or some other example(s) herein, wherein the first repetition comprises mapping three distinct data sets to a group of nine 26-tone RUs, with each data set repeated on three RUs, and the second repetition comprises mapping a single data set identically across the nine 26-tone RUs.
[0144] Example 21 may include the method of example 19 and/or some other example(s) herein, further comprising reuse predefined RU configurations comprising 26-tone, 52-tone, 106-tone, and 242-tone RUs to maintain backward compatibility.
[0145] Example 22 may include the method of example 19 and/or some other example(s) herein, wherein the repetition type may be determined based on a signal quality metric or a target range performance.
[0146] Example 23 may include the method of example 19 and/or some other example(s) herein, further comprising performing round-robin placement of repeated data using the plurality of 26-tone RUs to enhance frequency diversity gain.
[0147] Example 24 may include the method of example 19 and/or some other example(s) herein, wherein mapping the data may include assigning unique indices to each 26-tone RU of the plurality of 26-tone RUs.
[0148] Example 25 may include the method of example 19 and/or some other example(s) herein, further comprising encoding the data prior to mapping using a forward error correction code.
[0149] Example 26 may include the method of example 19 and/or some other example(s) herein, further comprising a transceiver configured to transmit and receive wireless signals.
[0150] Example 27 may include the method of example 26 and/or some other example(s) herein, further comprising an antenna coupled to the transceiver to cause to send the data.
[0151] Example 28 may include an apparatus comprising means for: receiving data to be transmitted over a wireless communication channel using a 20 MHz bandwidth and a plurality of 26-tone resource units (RUs); determining a repetition type for the plurality of 26-tone RUs based on a transmission configuration, wherein the repetition type may be selected from a first repetition and a second repetition; mapping the data to the plurality of 26-tone RUs based on the repetition type, wherein the first repetition maps different data sets to subsets of the plurality of 26-tone RUs, each repeated three times, and the second repetition maps identical data across the plurality of 26-tone RUs; and transmitting the data over the wireless communication channel using the plurality of 26-tone RUs.
[0152] Example 29 may include the apparatus of example 28 and/or some other example(s) herein, wherein the first repetition comprises mapping three distinct data sets to a group of nine 26-tone RUs, with each data set repeated on three RUs, and the second repetition comprises mapping a single data set identically across the nine 26-tone RUs.
[0153] Example 30 may include the apparatus of example 28 and/or some other example(s) herein, further comprising reuse predefined RU configurations comprising 26-tone, 52-tone, 106-tone, and 242-tone RUs to maintain backward compatibility.
[0154] Example 31 may include the apparatus of example 28 and/or some other example(s) herein, wherein the repetition type may be determined based on a signal quality metric or a target range performance.
[0155] Example 32 may include the apparatus of example 28 and/or some other example(s) herein, further comprising performing round-robin placement of repeated data using the plurality of 26-tone RUs to enhance frequency diversity gain.
[0156] Example 33 may include the apparatus of example 28 and/or some other example(s) herein, wherein mapping the data may include assigning unique indices to each 26-tone RU of the plurality of 26-tone RUs.
[0157] Example 34 may include the apparatus of example 28 and/or some other example(s) herein, further comprising encoding the data prior to mapping using a forward error correction code.
[0158] Example 35 may include the apparatus of example 28 and/or some other example(s) herein,

further comprising a transceiver configured to transmit and receive wireless signals.

[0159] Example 36 may include the apparatus of example 35 and/or some other example(s) herein, further comprising an antenna coupled to the transceiver to cause to send the data.

[0160] Example 37 may include one or more non-transitory computer-readable media comprising instructions to cause an electronic device, upon execution of the instructions by one or more processors of the electronic device, to perform one or more elements of a method described in or related to any of examples 1-36, or any other method or process described herein.

[0161] Example 38 may include an apparatus comprising logic, modules, and/or circuitry to perform one or more elements of a method described in or related to any of examples 1-36, or any other method or process described herein.

[0162] Example 39 may include a method, technique, or process as described in or related to any of examples 1-36, or portions or parts thereof.

[0163] Example 40 may include an apparatus comprising: one or more processors and one or more computer readable media comprising instructions that, when executed by the one or more processors, cause the one or more processors to perform the method, techniques, or process as described in or related to any of examples 1-36, or portions thereof.

[0164] Example 41 may include a method of communicating in a wireless network as shown and described herein.

[0165] Example 42 may include a system for providing wireless communication as shown and described herein.

[0166] Example 43 may include a device for providing wireless communication as shown and described herein.

[0167] Embodiments according to the disclosure are in particular disclosed in the attached claims directed to a method, a storage medium, a device and a computer program product, wherein any feature mentioned in one claim category, e.g., method, can be claimed in another claim category, e.g., system, as well. The dependencies or references back in the attached claims are chosen for formal reasons only. However, any subject matter resulting from a deliberate reference back to any previous claims (in particular multiple dependencies) can be claimed as well, so that any combination of claims and the features thereof are disclosed and can be claimed regardless of the dependencies chosen in the attached claims. The subject-matter which can be claimed comprises not only the combinations of features as set out in the attached claims but also any other combination of features in the claims, wherein each feature mentioned in the claims can be combined with any other feature or combination of other features in the claims. Furthermore, any of the embodiments and features described or depicted herein can be claimed in a separate claim and/or in any combination with any embodiment or feature described or depicted herein or with any of the features of the attached claims.

[0168] The foregoing description of one or more implementations provides illustration and description, but is not intended to be exhaustive or to limit the scope of embodiments to the precise form disclosed. Modifications and variations are possible in light of the above teachings or may be acquired from practice of various embodiments.

[0169] Certain aspects of the disclosure are described above with reference to block and flow diagrams of systems, methods, apparatuses, and/or computer program products according to various implementations. It will be understood that one or more blocks of the block diagrams and flow diagrams, and combinations of blocks in the block diagrams and the flow diagrams, respectively, may be implemented by computer-executable program instructions. Likewise, some blocks of the block diagrams and flow diagrams may not necessarily need to be performed in the order presented, or may not necessarily need to be performed at all, according to some implementations.

[0170] These computer-executable program instructions may be loaded onto a special-purpose computer or other particular machine, a processor, or other programmable data processing

apparatus to produce a particular machine, such that the instructions that execute on the computer, processor, or other programmable data processing apparatus create means for implementing one or more functions specified in the flow diagram block or blocks. These computer program instructions may also be stored in a computer-readable storage media or memory that may direct a computer or other programmable data processing apparatus to function in a particular manner, such that the instructions stored in the computer-readable storage media produce an article of manufacture including instruction means that implement one or more functions specified in the flow diagram block or blocks. As an example, certain implementations may provide for a computer program product, comprising a computer-readable storage medium having a computer-readable program code or program instructions implemented therein, said computer-readable program code adapted to be executed to implement one or more functions specified in the flow diagram block or blocks. The computer program instructions may also be loaded onto a computer or other programmable data processing apparatus to cause a series of operational elements or steps to be performed on the computer or other programmable apparatus to produce a computer-implemented process such that the instructions that execute on the computer or other programmable apparatus provide elements or steps for implementing the functions specified in the flow diagram block or blocks.
[0171] Accordingly, blocks of the block diagrams and flow diagrams support combinations of means for performing the specified functions, combinations of elements or steps for performing the specified functions and program instruction means for performing the specified functions. It will also be understood that each block of the block diagrams and flow diagrams, and combinations of blocks in the block diagrams and flow diagrams, may be implemented by special-purpose, hardware-based computer systems that perform the specified functions, elements or steps, or combinations of special-purpose hardware and computer instructions.
[0172] Conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain implementations could include, while other implementations do not include, certain features, elements, and/or operations. Thus, such conditional language is not generally intended to imply that features, elements, and/or operations are in any way required for one or more implementations or that one or more implementations necessarily include logic for deciding, with or without user input or prompting, whether these features, elements, and/or operations are included or are to be performed in any particular implementation.
[0173] Many modifications and other implementations of the disclosure set forth herein will be apparent having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the disclosure is not to be limited to the specific implementations disclosed and that modifications and other implementations are intended to be included within the scope of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A device, the device comprising processing circuitry coupled to storage, the processing circuitry configured to: receive data to be transmitted over a wireless communication channel using a 20 MHz bandwidth and a plurality of 26-tone resource units (RUs); determine a repetition type for the plurality of 26-tone RUs based on a transmission configuration, wherein the repetition type is selected from a first repetition and a second repetition; map the data to the plurality of 26-tone RUs based on the repetition type, wherein the first repetition maps different data sets to subsets of the plurality of 26-tone RUs, each repeated three times, and the second repetition maps identical data across the plurality of 26-tone RUs; and transmit the data over the wireless communication channel using the plurality of 26-tone RUs.
2. The device of claim 1, wherein the first repetition comprises mapping three distinct data sets to a

group of nine 26-tone RUs, with each data set repeated on three RUs, and the second repetition comprises mapping a single data set identically across the nine 26-tone RUs.

3. The device of claim 1, wherein the processing circuitry is further configured to reuse predefined RU configurations comprising 26-tone, 52-tone, 106-tone, and 242-tone RUs to maintain backward compatibility.

4. The device of claim 1, wherein the repetition type is determined based on a signal quality metric or a target range performance.

5. The device of claim 1, wherein the processing circuitry is further configured to perform round-robin placement of repeated data using the plurality of 26-tone RUs to enhance frequency diversity gain.

6. The device of claim 1, wherein mapping the data includes assigning unique indices to each 26-tone RU of the plurality of 26-tone RUs.

7. The device of claim 1, wherein the processing circuitry is further configured to encode the data prior to mapping using a forward error correction code.

8. The device of claim 1, further comprising a transceiver configured to transmit and receive wireless signals.

9. The device of claim 8, further comprising an antenna coupled to the transceiver to cause to send the data.

10. A non-transitory computer-readable medium storing computer-executable instructions which when executed by one or more processors result in performing operations comprising: receiving data to be transmitted over a wireless communication channel using a 20 MHz bandwidth and a plurality of 26-tone resource units (RUs); determining a repetition type for the plurality of 26-tone RUs based on a transmission configuration, wherein the repetition type is selected from a first repetition and a second repetition; mapping the data to the plurality of 26-tone RUs based on the repetition type, wherein the first repetition maps different data sets to subsets of the plurality of 26-tone RUs, each repeated three times, and the second repetition maps identical data across the plurality of 26-tone RUs; and transmitting the data over the wireless communication channel using the plurality of 26-tone RUs.

11. The non-transitory computer-readable medium of claim 10, wherein the first repetition comprises mapping three distinct data sets to a group of nine 26-tone RUs, with each data set repeated on three RUs, and the second repetition comprises mapping a single data set identically across the nine 26-tone RUs.

12. The non-transitory computer-readable medium of claim 10, wherein the operations further comprise reuse predefined RU configurations comprising 26-tone, 52-tone, 106-tone, and 242-tone RUs to maintain backward compatibility.

13. The non-transitory computer-readable medium of claim 10, wherein the repetition type is determined based on a signal quality metric or a target range performance.

14. The non-transitory computer-readable medium of claim 10, wherein the operations further comprise performing round-robin placement of repeated data using the plurality of 26-tone RUs to enhance frequency diversity gain.

15. The non-transitory computer-readable medium of claim 10, wherein mapping the data includes assigning unique indices to each 26-tone RU of the plurality of 26-tone RUs.

16. The non-transitory computer-readable medium of claim 10, wherein the operations further comprise encoding the data prior to mapping using a forward error correction code.

17. The non-transitory computer-readable medium of claim 10, further comprising a transceiver configured to transmit and receive wireless signals.

18. The non-transitory computer-readable medium of claim 17, further comprising an antenna coupled to the transceiver to cause to send the data.

19. A method comprising: receiving data to be transmitted over a wireless communication channel using a 20 MHz bandwidth and a plurality of 26-tone resource units (RUs); determining a repetition type for the plurality of 26-tone RUs based on a transmission configuration, wherein the repetition type is selected from a first repetition and a second repetition; mapping the data to the plurality of 26-tone RUs based on the repetition type, wherein the first repetition maps different data sets to subsets of the plurality of 26-tone RUs, each repeated three times, and the second repetition maps identical data across the plurality of 26-tone RUs; and transmitting the data over the wireless communication channel using the plurality of 26-tone RUs.

20. The method of claim 19, wherein the first repetition comprises mapping three distinct data sets to a group of nine 26-tone RUs, with each data set repeated on three RUs, and the second repetition comprises mapping a single data set identically across the nine 26-tone RUs.